

The Mathematics of Rough Volatility: From Fractional Brownian Motion to the Hurst Exponent

Marc Amargós

Abstract

Recent work has shown that volatility in financial markets behaves as a fractional Brownian motion with Hurst exponent $H \approx 0.1$, making it far rougher than the classical assumption of Brownian-like smoothness. This paper's objective is to develop the mathematical foundation underlying that claim. Starting from the four defining properties of standard Brownian motion, we construct fractional Brownian motion as a one-parameter generalization indexed by $H \in (0, 1)$. Then we derive its increment-correlation structure and the three resulting regimes of path behavior. After this, we use its moment-scaling property to build an operational estimator of H based on log-log regression. This estimator is validated against exact Davies and Harte (1987) simulations across four values of H spanning all three regimes, recovering the true parameter with bias no greater than 0.002 and standard deviation below 0.02 across 100 replications per value. This confidence in the estimator has been established through simulation and is what allows it to be applied to real volatility data with real interpretive weight. It also lays the groundwork for the empirical estimation of roughness in real volatility data.

Keywords: Rough volatility; Fractional Brownian motion; Hurst exponent; Self-similarity; Long-range dependence; Moment scaling; Monte Carlo simulation

Introduction

For decades, the standard modeling assumption in mathematical finance was that stock market volatility evolves smoothly and its trajectory resembles at worst an ordinary Brownian motion. This assumption was upended in 2018. Gatheral et al. (2018) used high-frequency realized volatility data across a range of equity indices to demonstrate, empirically, that log-volatility behaves as a fractional Brownian motion with Hurst exponent $H \approx 0.1$. This is an order of magnitude rougher than Brownian motion itself, and in the opposite regime ($H < \frac{1}{2}$) from the long-memory models that had dominated the literature since Comte and Renault (1998). This single empirical finding, now known as “rough volatility”, reshaped how practitioners think about volatility dynamics, option pricing near expiry, and the notion of “memory” in financial time series.

The claim that volatility has $H \approx 0.1$ is often repeated without clarifying what H actually is. Many questions arise such as: why it can be estimated from data at all, or what mathematical structure makes such an estimate meaningful in the first place. We intend to fill that gap. The following work does not aim to present new empirical findings. Its contribution is to build a clear mathematical path to understanding what “rough volatility” means and how one would go about testing for it. It is aimed at a reader who is comfortable with basic probability but has not necessarily encountered fractional Brownian motion before. We build this path from the four defining properties of standard Brownian motion to a fully operational, simulation-validated estimator of H .

The paper proceeds as follows. Section 1 reviews standard Brownian motion. Its objective is to establish its four defining properties, covariance function, and scaling property as the benchmark against which all subsequent generalization is measured. Section 2 introduces fractional Brownian motion as a one-parameter generalization of this benchmark, indexed by the Hurst exponent $H \in (0, 1)$, and shows that it reduces exactly to the Brownian case at $H = \frac{1}{2}$. Section 3 derives the correlation structure of fBM increments. This section identifies the three regimes, anti-persistent, neutral, and persistent, that H governs, and connects this to Hölder regularity of sample paths. Section 4 derives a moment-scaling relation from this theory and converts it into an operational log-log regression estimator of H . Section 5 validates that estimator against exact simulations with known ground truth. It confirms that it recovers the true H with negligible bias across all three regimes.

1 Standard Brownian Motion

Brownian motion first appeared in 1827, when botanist Robert Brown observed erratic jittering of pollen grains suspended in water. That phenomenon was later explained in more detail by Albert Einstein in 1905 in terms of molecular collisions. Louis Bachelier had already proposed the same process in 1900, but this time as a model for stock price fluctuations. He explained his findings in his thesis *Théorie de la spéculation*, the first mathematical model to be used for financial markets. The process was later put on a rigorous mathematical footing in the 1920s by Norbert Wiener, who proved the existence of a continuous-time random process with the properties physicists and economists had assumed. Hence the name Wiener process. Brownian motion is the canonical building block of continuous-time stochastic modeling. It underpinned the Itô calculus and workhorse models such as Black–Scholes. Brownian motion is regarded as the “neutral” and memoryless benchmark case, corresponding to Hurst exponent $H = \frac{1}{2}$. The neutrality offers the reference point against which the rough or non-Markovian behavior of fractional Brownian motion can be understood.

Definition 1.1 (Standard Brownian Motion). *A standard Brownian motion is a stochastic process $\{W(t), t \geq 0\}$ satisfying:*

- (i) $W(0) = 0$ almost surely;
- (ii) **Independent increments:** for any $0 \leq t_1 < t_2 \leq t_3 < t_4$, the increments $W(t_2) - W(t_1)$ and $W(t_4) - W(t_3)$ are independent;
- (iii) **Gaussian stationary increments:** $W(t) - W(s) \sim N(0, t - s)$ for any $0 \leq s < t$;
- (iv) **Continuous paths:** $t \mapsto W(t)$ is continuous almost surely.

Property (i) defines the starting point of the process at the origin. It is a normalization convention that has no substantive content of its own. Property (ii) captures the “memoryless” feature of Brownian motion. The events over a future time window are not affected by the events over a disjoint past window. Property (iii) affirms that the size of a move over an interval of length $t - s$ is normally distributed, with variance growing linearly in the elapsed time. Consequently, the typical magnitude of a move scales as the square root of $t - s$. This is the “square-root-of-time” rule, and since the distribution depends only on the length of the interval and not on when it occurs, it reflects the time-homogeneity of the process. Property (iv) states that sample paths have no jumps. However, as we demonstrate below, they are nevertheless nowhere differentiable. It constitutes the first idea of roughness this paper will generalize and control via the Hurst parameter H .

Proposition 1.2 (Covariance Function). *For all $s, t \geq 0$,*

$$\text{Cov}(W(s), W(t)) = \min(s, t). \quad (1)$$

Proof. Without loss of generality, assume $s < t$. The case $s = t$ is trivial and the case $s > t$ follows by symmetry. Write

$$W(t) = W(s) + (W(t) - W(s)).$$

Then

$$\text{Cov}(W(s), W(t)) = \text{Cov}(W(s), W(s)) + \text{Cov}(W(s), W(t) - W(s)).$$

The first term equals $\text{Var}(W(s)) = s$ by Property (iii) of Definition 1.1 applied to the interval $[0, s]$. The second term is zero since $W(s) = W(s) - W(0)$ and $W(t) - W(s)$ are increments over the non-overlapping intervals $[0, s]$ and $[s, t]$, and are therefore independent by Property (ii), hence uncorrelated. It follows that

$$\text{Cov}(W(s), W(t)) = s = \min(s, t). \quad \square$$

Proposition 1.3 (Scaling Property). *For any $c > 0$, the process $\{W(ct), t \geq 0\}$ has the same law as $\{\sqrt{c} W(t), t \geq 0\}$; that is,*

$$W(ct) \stackrel{d}{=} \sqrt{c} W(t). \quad (2)$$

Proof. Both processes are centered Gaussian, so equality in law is equivalent to equality of covariance functions. For $s \leq t$,

$$\text{Cov}(W(cs), W(ct)) = \min(cs, ct) = c \min(s, t) = \text{Cov}(\sqrt{c} W(s), \sqrt{c} W(t)). \quad \square$$

It can be regarded as a first instance of self-similarity: rescaling time by c is statistically equivalent to rescaling the process itself by $\sqrt{c} = c^{1/2}$. The exponent $\frac{1}{2}$ is the value taken, in the Brownian case, by a more general parameter: the Hurst parameter H . Generalizing to $H \neq \frac{1}{2}$ gives rise to fractional Brownian motion.

By property (iv), the sample paths of W are continuous everywhere. Nevertheless, a classical and counterintuitive fact is that these same paths are, with probability one, differentiable nowhere. In light of this, Brownian motion is considered “rough” at every scale, even in the absence of jumps. Whether this roughness is a fixed feature of the process, or whether it can be tuned, continuously, to be either smoother or rougher than the Brownian benchmark raises a natural question. Section 2 introduces fractional Brownian motion as precisely this generalization.

2 Fractional Brownian Motion

Fractional Brownian motion, fBM, introduces a single tunable parameter which is the Hurst exponent $H \in (0, 1)$. The exponent controls the degree of dependence between increments and, with it, the smoothness or roughness of the resulting sample paths. Standard Brownian motion is recovered as the special and memoryless case $H = \frac{1}{2}$, so that fBM can be understood not as a departure from Section 1, but as its natural generalization.

Definition 2.1 (Fractional Brownian Motion). *For a fixed Hurst parameter $H \in (0, 1)$, a fractional Brownian motion $\{B_H(t), t \geq 0\}$ is the centered Gaussian process with $B_H(0) = 0$ and covariance function*

$$E[B_H(t)B_H(s)] = \frac{1}{2} (t^{2H} + s^{2H} - |t - s|^{2H}), \quad s, t \geq 0, \quad (3)$$

whose sample paths $t \mapsto B_H(t)$ are continuous almost surely.

Proposition 2.2 (Reduction to Standard Brownian Motion). *For $H = \frac{1}{2}$, the fractional Brownian motion covariance function reduces to*

$$E[B_{1/2}(t)B_{1/2}(s)] = \min(s, t), \quad (4)$$

so that $B_{1/2}$ is a standard Brownian motion in the sense of Definition 1.1.

Proof. Setting $H = \frac{1}{2}$ in the covariance formula of Definition 2.1 gives $2H = 1$, so

$$E[B_{1/2}(t)B_{1/2}(s)] = \frac{1}{2} (t + s - |t - s|).$$

Assume without loss of generality that $s \leq t$, so that $|t - s| = t - s$. Substituting,

$$\frac{1}{2} (t + s - (t - s)) = \frac{1}{2} (2s) = s = \min(s, t).$$

This matches the covariance function established in Proposition 1.2, confirming that $B_{1/2}$ coincides in law with standard Brownian motion. $\square$

Proposition 2.3 (Stationary Increments). *For all $s, t \geq 0$,*

$$\text{Var}(B_H(t) - B_H(s)) = |t - s|^{2H}. \quad (5)$$

Proof. Expand the variance of the increment as

$$E[(B_H(t) - B_H(s))^2] = E[B_H(t)^2] + E[B_H(s)^2] - 2E[B_H(t)B_H(s)].$$

By Definition 2.1, setting $s = t$ in the covariance formula gives

$$E[B_H(t)^2] = \frac{1}{2} (t^{2H} + t^{2H} - 0) = t^{2H},$$

and similarly $E[B_H(s)^2] = s^{2H}$. Substituting these together with the covariance $E[B_H(t)B_H(s)] = \frac{1}{2} (t^{2H} + s^{2H} - |t - s|^{2H})$ from Definition 2.1,

$$E[(B_H(t) - B_H(s))^2] = t^{2H} + s^{2H} - 2 \cdot \frac{1}{2} (t^{2H} + s^{2H} - |t - s|^{2H}) = |t - s|^{2H}.$$

Since B_H is centered ($E[B_H(t) - B_H(s)] = 0$), this second moment coincides with the variance, giving $\text{Var}(B_H(t) - B_H(s)) = |t - s|^{2H}$. $\square$

Proposition 2.4 (Self-Similarity). *For any $c > 0$, the process $\{B_H(ct), t \geq 0\}$ has the same law as $\{c^H B_H(t), t \geq 0\}$; that is,*

$$B_H(ct) \stackrel{d}{=} c^H B_H(t). \quad (6)$$

Proof. As in the proof of Proposition 1.3, both sides are centered Gaussian processes, so it suffices to match covariance functions. Using Definition 2.1,

$$\begin{aligned} E[B_H(ct)B_H(cs)] &= \frac{1}{2} ((ct)^{2H} + (cs)^{2H} - |ct - cs|^{2H}) \\ &= c^{2H} \cdot \frac{1}{2} (t^{2H} + s^{2H} - |t - s|^{2H}) \\ &= c^{2H} E[B_H(t)B_H(s)], \end{aligned}$$

which is exactly the covariance of $c^H B_H(t)$, since $E[c^H B_H(t) \cdot c^H B_H(s)] = c^{2H} E[B_H(t)B_H(s)]$. As the two processes agree in every finite-dimensional covariance, they agree in law. $\square$

This is the self-similarity established for standard Brownian motion in Proposition 1.3. The scaling exponent is generalized from the fixed value $\frac{1}{2}$ to the tunable Hurst parameter H . The exponent H plays the role that $\frac{1}{2}$ played before.

Unlike standard Brownian motion, however, the increments of B_H are not independent for $H \neq \frac{1}{2}$. Property (ii) of Definition 1.1, independence of increments over disjoint intervals, was a defining feature of the Brownian case, but for general H it fails. The increments of fBM are correlated and both the sign and the strength of this correlation are governed by the Hurst parameter H . The correlation can be positive, giving rise to trend-reinforcing and persistent paths. It can also be negative, giving rise to trend-reversing and anti-persistent paths. Whether this correlation is positive or negative, and precisely how this dependence is encoded in H , is the question taken up in Section 3.

3 The Hurst Parameter and Its Three Regimes

The increments of fractional Brownian motion remain stationary for every H , but they are not independent when $H \neq \frac{1}{2}$. The following section derives an explicit formula for the correlation between non-overlapping increments, showing that its sign and magnitude are governed entirely by the Hurst parameter H . Three distinct regimes of path behavior then emerge, isolating in particular the regime $H < \frac{1}{2}$ that turns out to be empirically relevant for financial volatility.

Proposition 3.1 (Correlation of Consecutive Unit Increments). *Define $\rho_H = \text{Corr}(B_H(1) - B_H(0), B_H(2) - B_H(1))$. Then*

$$\rho_H = 2^{2H-1} - 1. \quad (7)$$

Proof. Since $B_H(0) = 0$, write $X = B_H(1) - B_H(0) = B_H(1)$ and $Y = B_H(2) - B_H(1)$.

Variances. By Proposition 2.3, $\text{Var}(X) = |1 - 0|^{2H} = 1$ and $\text{Var}(Y) = |2 - 1|^{2H} = 1$.

Covariance. By bilinearity,

$$\text{Cov}(X, Y) = \text{Cov}(B_H(1), B_H(2) - B_H(1)) = E[B_H(1)B_H(2)] - E[B_H(1)^2].$$

Using Definition 2.1 with $t = 2$, $s = 1$:

$$E[B_H(1)B_H(2)] = \frac{1}{2} (2^{2H} + 1^{2H} - |2 - 1|^{2H}) = \frac{1}{2} (2^{2H} + 1 - 1) = \frac{1}{2} \cdot 2^{2H} = 2^{2H-1}.$$

And from Step 2 of the earlier derivation, $E[B_H(1)^2] = 1^{2H} = 1$. Hence $\text{Cov}(X, Y) = 2^{2H-1} - 1$.

Correlation. Since $\text{Var}(X) = \text{Var}(Y) = 1$,

$$\rho_H = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \text{Cov}(X, Y) = 2^{2H-1} - 1. \quad \square$$

Proposition 3.1 shows that the sign of $\rho_H = 2^{2H-1} - 1$ is determined entirely by how H compares to $\frac{1}{2}$ since $2^{2H-1} - 1$ is an increasing function of H that vanishes precisely at $H = \frac{1}{2}$. The Hurst parameter is hence divided into three distinct regimes.

When $H < \frac{1}{2}$, we have $\rho_H < 0$. Increments are anti-persistent, meaning a positive move tends to be followed by a negative one, and vice versa. It is as though the process is constantly correcting itself. We observe a mean-reverting tendency at the level of increments that produces sample paths that are rougher and irregular at every timescale compared to standard Brownian motion.

When $H = \frac{1}{2}$, we have $\rho_H = 0$. Consecutive increments are not correlated. They also recover the independence that defines standard Brownian motion. This is the neutral case, memoryless in either direction, and it is exactly as established in Proposition 2.2.

When $H > \frac{1}{2}$, we have $\rho_H > 0$ and increments are persistent. We deduce that a positive move tends to be followed by another positive move, producing long-range dependence and trending behavior. It also generates sample paths that are smoother-looking than Brownian motion. This is the regime originally proposed for volatility by Comte and Renault (1998).

Letting $H \rightarrow \frac{1}{2}$ in the formula, as a sanity check, gives $\rho_H \rightarrow 2^0 - 1 = 0$. It correctly recovers the independence result of Proposition 2.2, and confirms that $H = \frac{1}{2}$ is not an arbitrary boundary but the unique value at which the correlation formula itself forces independence.

For $H > \frac{1}{2}$, this positive correlation is not only present but strong enough to be economically significant. In fact, the autocovariances of the increments decay so slowly as the lag grows that they fail to be summable. That means

$$\sum_{k=1}^{\infty} \text{Cov}(B_H(1), B_H(k+1) - B_H(k)) = \infty. \quad (8)$$

It is the property known as long-range dependence (Beran, 2017; Nourdin et al., 2012), and it is what motivated the original choice of $H > \frac{1}{2}$ for modeling volatility persistence in Comte and Renault (1998).

The intuitive language of “roughness” used throughout this section can be made mathematically precise via Hölder regularity. It is a classical result (see, e.g., (Mandelbrot and Van Ness, 1968; Nourdin et al., 2012)) that the sample paths of B_H are almost surely Hölder continuous of any order $\gamma < H$, but not of any order $\gamma > H$. Concretely, this means that for every $\gamma < H$, there exists (locally) a constant C such that $|B_H(t) - B_H(s)| \leq C|t - s|^\gamma$ for all s, t in a given interval, while no such bound holds once γ exceeds H . It explains why H can be treated as a direct “roughness

dial". A smaller H forces the Hölder exponent down. It allows the path to oscillate more sharply at small scales. On the other hand, a larger H permits a higher exponent and correspondingly smoother-looking trajectories, with $H = \frac{1}{2}$ recovering the familiar Hölder- $(\frac{1}{2} - \varepsilon)$ regularity of standard Brownian motion.

The financial relevance of this framework was established empirically by Gatheral et al. (2018). They estimated H directly from high-frequency realized volatility data across a range of equity indices and found $H \approx 0.1$. It was squarely in the anti-persistent, rough regime, and far below the $H > \frac{1}{2}$ long-range-dependence assumption that had dominated the volatility modeling literature since Comte and Renault (1998). The finding overturns a long-standing stylized fact: instead of being smooth and slowly evolving, volatility is far rougher than even standard Brownian motion, let alone the persistent processes classical models assume. What remains is a question the theory developed so far cannot answer on its own, given only observed price data: how does one actually estimate H ?

4 Moment Scaling and Estimation of the Hurst Parameter

Sections 2 and 3 characterized fractional Brownian motion through the single parameter H , which governs both the correlation structure of increments and the Hölder regularity of sample paths. The inverse problem remains: given only an observed, discretely-sampled realization of a process, how does one recover an estimate of H from the data itself?

Proposition 4.1 (Moment Scaling Relation). *For any $q > 0$,*

$$E[|B_H(t + \Delta) - B_H(t)|^q] = K_q \Delta^{qH}, \quad K_q = E[|Z|^q], \quad Z \sim N(0, 1). \quad (9)$$

Proof. By Proposition 2.3, the increment $B_H(t + \Delta) - B_H(t)$ is normally distributed with mean zero and variance Δ^{2H} , i.e. $B_H(t + \Delta) - B_H(t) \sim N(0, \Delta^{2H})$. Any such Gaussian random variable can be written in distribution as its standard deviation times a standard normal variable:

$$B_H(t + \Delta) - B_H(t) \stackrel{d}{=} \Delta^H Z, \quad Z \sim N(0, 1).$$

Raising both sides to the power q in the absolute value and taking expectations,

$$E[|B_H(t + \Delta) - B_H(t)|^q] = E[|\Delta^H Z|^q] = \Delta^{qH} E[|Z|^q],$$

where Δ^{qH} is deterministic and factors outside the expectation. Setting $K_q := E[|Z|^q]$ gives the claimed relation. $\square$

Remark. The constant K_q admits a closed form via the standard absolute moments of the Gaussian distribution:

$$K_q = \frac{2^{q/2} \Gamma\left(\frac{q+1}{2}\right)}{\sqrt{\pi}} \quad (10)$$

(see, e.g., Winkelbauer (2012), for a derivation of general absolute Gaussian moments).

4.1 The log-log linearization

Taking logarithms of both sides of Proposition 4.1 turns the power-law relation into a linear one:

$$\log E[|B_H(t + \Delta) - B_H(t)|^q] = \log K_q + qH \cdot \log \Delta. \quad (11)$$

This is the operational core of the estimation procedure. If one computes the empirical q -th absolute moment of the increments at several different lags Δ , and plots the logarithm of this empirical moment against $\log \Delta$, the theory predicts the points should lie on a straight line. The slope must be exactly qH and the intercept $\log K_q$. The intercept plays no role in recovering H and can be treated as a nuisance parameter. Hence the entire estimation problem reduces to a single quantity, the slope of this line. Given an estimated slope $\hat{\beta}$ from ordinary least squares regression of $\log m(q, \Delta)$ on $\log \Delta$, the estimator of the Hurst parameter is simply

$$\hat{H} = \frac{\hat{\beta}}{q}. \quad (12)$$

4.2 The empirical estimator, made precise

In practice, one observes a discretely-sampled path $X_0, X_\delta, X_{2\delta}, \dots, X_{N\delta}$ over some fixed mesh δ , rather than the continuous process itself. For a given lag Δ (a multiple of δ), define the empirical q -th absolute moment as the sample average of q -th powers of the observed increments:

$$m(q, \Delta) = \frac{1}{N_\Delta} \sum_t |X_{t+\Delta} - X_t|^q, \quad (13)$$

where the sum runs over all available (overlapping) pairs of observations spaced Δ apart, i.e. $t = 0, \delta, 2\delta, \dots$; N_Δ is the number of such pairs. By Proposition 4.1 and the linearization of Section 4.1, $m(q, \Delta)$ is the natural empirical counterpart of $E[|B_H(t + \Delta) - B_H(t)|^q]$, so that $\log m(q, \Delta)$ should be approximately linear in $\log \Delta$ with slope $\zeta_q := qH$.

The estimation procedure is then as follows. For a fixed q , compute $m(q, \Delta)$ across a range of lags Δ , then regress $\log m(q, \Delta)$ on $\log \Delta$ by ordinary least squares, taking the fitted slope as an estimate $\hat{\zeta}_q$ of qH . Dividing by q gives $\hat{H} = \hat{\zeta}_q/q$. Repeating this across several values of q (typically $q = 0.5, 1, 1.5, 2, 3$) yields a set of slope estimates $\hat{\zeta}_q$ that, under the fBM hypothesis, should themselves be linear in q with slope H and zero intercept. That is, $\zeta_q = qH$ is a genuine testable prediction of the model, not merely a computational device. The diagnostic above is employed by Gatheral et al. (2018) on realized volatility data, where the near-perfect linearity of ζ_q against q across multiple equity indices served as direct empirical support for the fBM specification, well before any single point estimate of H was reported.

4.3 Estimation caveats

This estimator is operationally simple but contains multiple subtleties. The choice of lag range trades off two competing sources of error. Lags too small are dominated by microstructure noise, while lags too large leave too few overlapping observations N_Δ to estimate $m(q, \Delta)$ reliably, inflating its variance. Higher values of q compound this problem further, since raising increments to a larger power amplifies the influence of tail observations and outliers on the empirical moment. There is also a subtler statistical issue. Because $\log$ is a concave function, Jensen's inequality implies that $E[\log m(q, \Delta)] \leq \log E[m(q, \Delta)]$. The expected value of the log-transformed empirical moment understates the log of the true moment it is estimating. Therefore the regression estimator carries a small, generally negative finite-sample bias in $\hat{H}$. These issues are discussed in more detail by Gatheral et al. (2018). While they do not undermine the estimator's validity, they are worth bearing in mind when interpreting point estimates from limited samples rather than treating $\hat{H}$ as an exact and noise-free quantity.

The natural validation step is to simulate paths of fractional Brownian motion with a known H and check whether the estimator of Section 4.2 recovers it.

5 Simulation and Numerical Validation of the Estimator

Before applying the estimator of Section 4 to real market data, it must be validated in a setting where the ground truth is known: do the theoretical results of Sections 2–4 hold when H is chosen in advance and paths are simulated accordingly?

5.1 Experiment design

Sample paths were simulated using the Davies and Harte (1987) method. This is an exact circulant-embedding algorithm that constructs a Gaussian vector with the target fBM covariance. It does so by embedding it into a larger circulant matrix and exploiting the Fast Fourier Transform for efficient, exact sampling (Davies and Harte, 1987; Dieker, 2004). Unlike approximate simulation schemes, Davies and Harte (1987) produces paths with the exact finite-dimensional covariance structure of Definition 2.1. It makes it the appropriate benchmark for validating an estimator built on the same covariance. Simulation was carried out in Python using the `fbm` package, which implements this algorithm directly.

Four values of the Hurst parameter were considered, spanning the three regimes identified in Section 3: $H \in \{0.1, 0.3, 0.5, 0.7\}$ (rough, mildly rough, neutral, and persistent, respectively). For each value of H , 100 independent paths of length $N = 10,000$ were simulated. For every path, the estimator of Section 4.2 was applied: the empirical moment $m(q, \Delta)$ was computed for $q \in \{0.5, 1, 1.5, 2, 3\}$ across lags $\Delta = 1, \dots, 50$, $\log m(q, \Delta)$ was regressed on $\log \Delta$ to obtain $\hat{\zeta}_q$, and the resulting per- q estimates $\hat{\zeta}_q/q$ were averaged to produce a single $\hat{H}$ per path. Reporting the mean and standard deviation of $\hat{H}$ across the 100 replications, rather than a single realization, avoids drawing conclusions from what could otherwise be an atypically lucky or unlucky draw.

5.2 Results

The estimator was applied to 100 independent replications at each value of H . The results are summarized in Table 1.

H_{true}	Mean $\hat{H}$	Std. dev.
0.10	0.099	0.005
0.30	0.299	0.011
0.50	0.500	0.014
0.70	0.698	0.017

Table 1: Estimation results across 100 replications per value of H , paths of length $N = 10,000$.

Across all four regimes, the estimator recovers the true H with a bias of no more than 0.002 in absolute value and a standard deviation that remains under 0.02 even at $N = 10,000$. There is a mild tendency for the standard deviation to increase with H , consistent with the fact that smoother or more persistent paths carry less independent information per observation. But this does not translate into any economically meaningful degradation in accuracy at this sample size.

Figure 1 shows one simulated sample path for each of the four H values, generated from a common random seed. The visual contrast is stark. The $H = 0.1$ path is markedly more jagged than the $H = 0.5$ benchmark, while the $H = 0.7$ path visibly trends and looks noticeably smoother, precisely the roughness ordering established theoretically in Section 3, now visible to the eye.

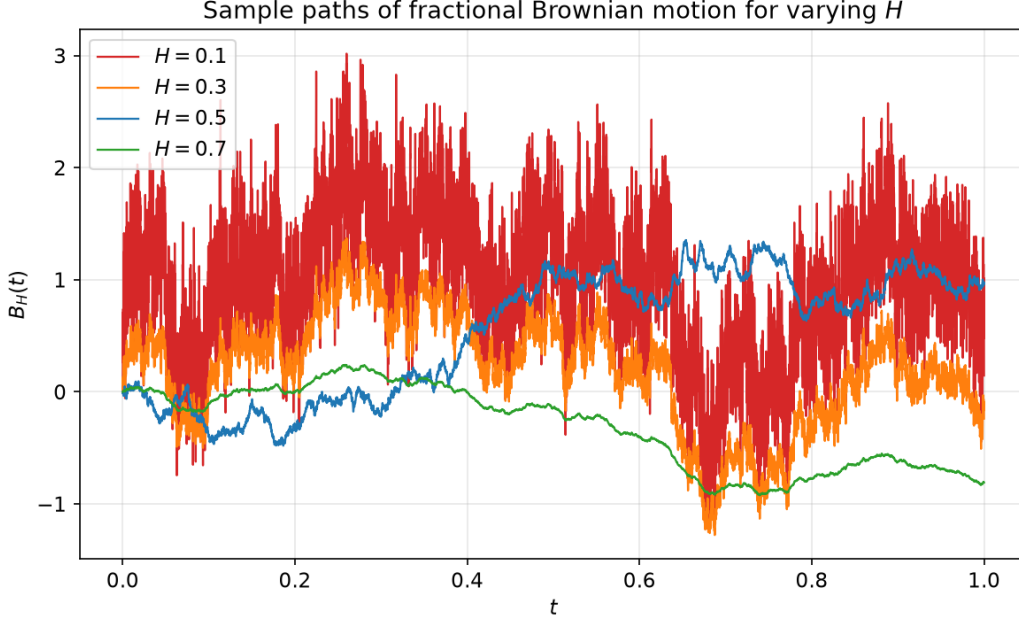


Figure 1: Sample paths of fractional Brownian motion for $H \in \{0.1, 0.3, 0.5, 0.7\}$.

Figure 2 illustrates the regression mechanics of Section 4.1 directly, plotting $\log m(q, \Delta)$ against $\log \Delta$ for a single simulated path with $H_{\text{true}} = 0.3$ and $q = 2$. The empirical points lie almost exactly on the fitted line, with an estimated slope of $\hat{\zeta}_2 = 0.597$, giving $\hat{H} = 0.299$. It is a striking confirmation, on a single path, of the linear relationship predicted by Proposition 4.1. A closer look at the largest lags in Figure 2 ($\log \Delta > 3.5$) reveals a slight downward deviation from the fitted line. This occurs because at large Δ , the number of available increments $N - \Delta$ shrinks and the remaining increments overlap more heavily. Therefore the empirical moment is computed from far fewer effectively independent observations, noisier, and systematically more affected by the concavity of the logarithm. This is why the lag range was capped at $\Delta = 50 = N/200$.

Figure 3 performs the second-order diagnostic proposed in Section 4.2: plotting the fitted slopes $\hat{\zeta}_q$ against q itself, for the same path used in Figure 2. The theory predicts $\zeta_q = qH$, a line through the origin with slope H ; the fitted slope of $\hat{\zeta}_q$ against q here is 0.298, matching the true $H = 0.3$ almost exactly. This diagnostic is identical to the one later used by Gatheral et al. (2018) on realized volatility data. Its success here on data with a known, controlled ground truth confirms that the linearity of ζ_q in q is a verifiable signature of the fBM model rather than an artifact of the estimation procedure.

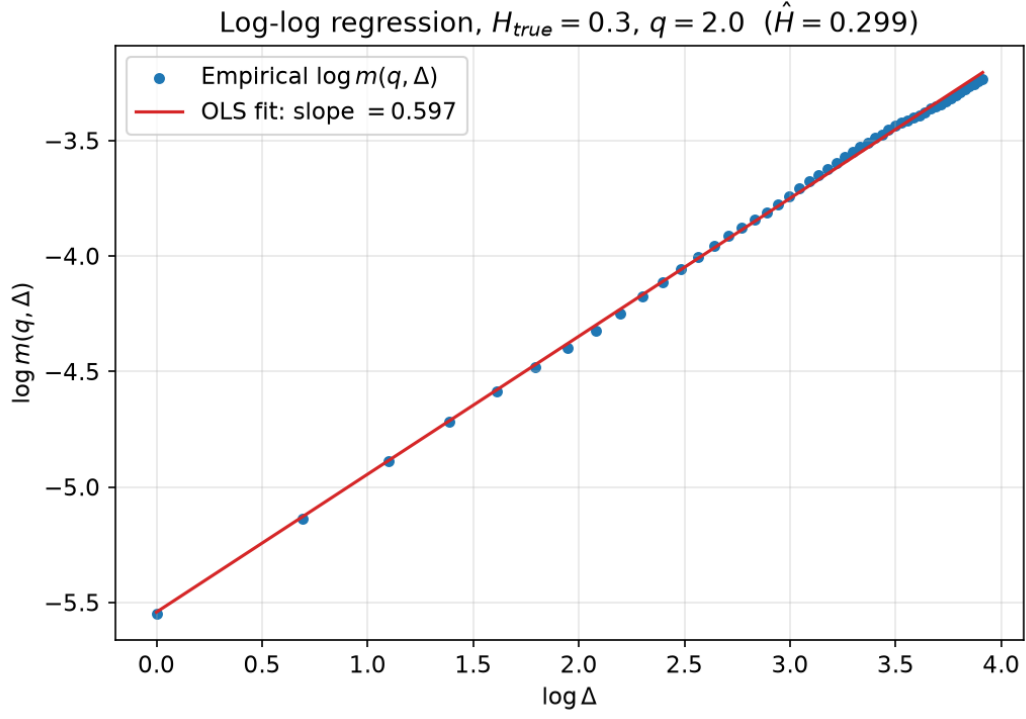


Figure 2: Regression of $\log m(q, \Delta)$ on $\log \Delta$ for $H_{true} = 0.3$, $q = 2$.

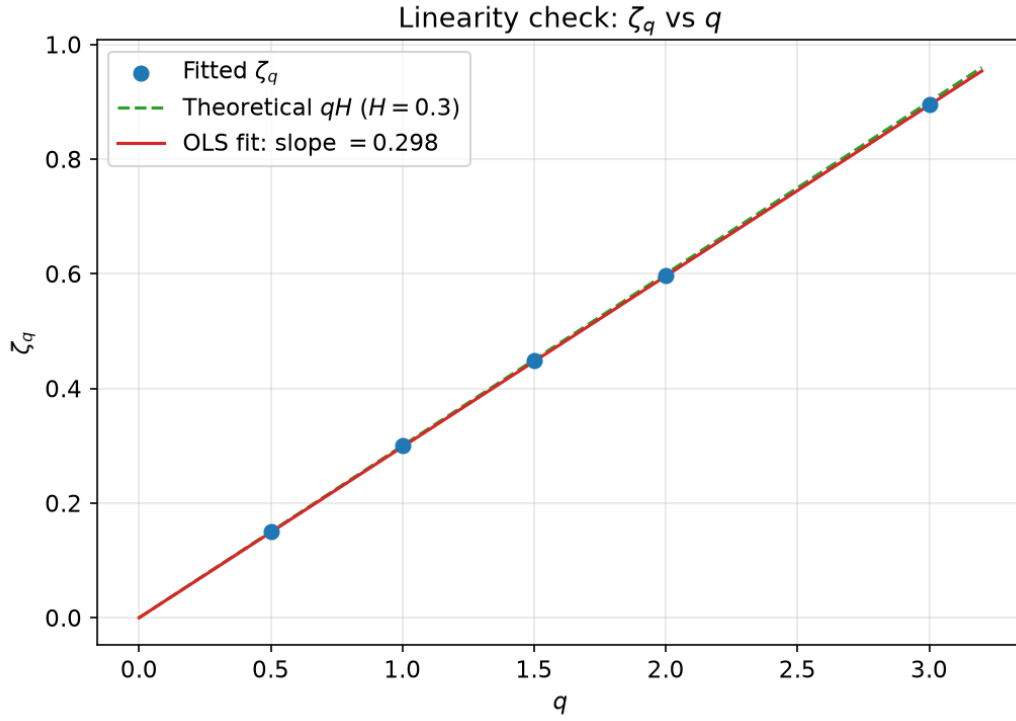


Figure 3: Linearity check: fitted $\hat{\zeta}_q$ against q , compared with the theoretical line $\zeta_q = qH$.

5.3 Discussion and limitations

The results confirm that the estimator of Section 4 performs well under ideal conditions. That means exact simulation, no microstructure noise, and a sample size ($N = 10,000$) considerably larger than what a typical daily-frequency financial dataset would provide. The caveats flagged in Section 4.3 are only mildly visible here. In fact, the observed bias is small and consistently negative, as the concavity of the logarithm predicts, but at this sample size it is dwarfed by sampling variability. With 100 replications, the standard error of the mean $\hat{H}$ is $\text{std}/\sqrt{100}$. For $H = 0.1$ this is approximately 0.0005, so the observed bias of -0.001 is roughly two standard errors from zero. It is borderline but suggestive of a small systematic effect rather than pure sampling noise, a distinction that additional replications would sharpen further. Whether this bias becomes more pronounced at smaller sample sizes, shorter lag ranges, or under the microstructure noise present in real high-frequency data is a question this controlled experiment cannot answer. It is the starting point for a planned companion paper applying the estimator to real volatility series.

Conclusion

We have built, from first principles, the mathematical machinery needed to make precise a single question: how rough is volatility? Starting from the four defining properties of standard Brownian motion, we constructed fractional Brownian motion as its natural one-parameter generalization. We then showed that it reduces exactly to the Brownian case at $H = \frac{1}{2}$, and derived its increment-correlation structure to identify three distinct regimes. These regimes are anti-persistent, neutral, and persistent, governed by the Hurst exponent H . From this theoretical foundation, we derived a moment-scaling relation linking H to the growth rate of absolute increment moments. We converted it into an operational log-log regression estimator, and validated that estimator against exact Davies and Harte (1987) simulations across all three regimes. This allowed us to recover the true H with negligible bias and tight standard deviation at large sample sizes.

Because H can be recovered from data through nothing more than a linear regression on logged moments, “rough volatility” is not merely a theoretical construct but a hypothesis that can be tested directly against real financial time series.

References

- Jan Beran. *Statistics for long-memory processes*. Routledge, 2017.
- Fabienne Comte and Eric Renault. Long memory in continuous-time stochastic volatility models. *Mathematical finance*, 8(4):291–323, 1998.
- Robert B Davies and David S Harte. Tests for hurst effect. *Biometrika*, 74(1):95–101, 1987.
- Ton Dieker. Simulation of fractional brownian motion. Master’s thesis, University of Twente, 2004.
- Jim Gatheral, Thibault Jaisson, and Mathieu Rosenbaum. Volatility is rough. *Quantitative Finance*, 18(6):933–949, 2018.
- Benoit B Mandelbrot and John W Van Ness. Fractional brownian motions, fractional noises and applications. *SIAM review*, 10(4):422–437, 1968.
- Ivan Nourdin et al. *Selected aspects of fractional Brownian motion*, volume 4. Springer, 2012.

Andreas Winkelbauer. Moments and absolute moments of the normal distribution. *arXiv preprint arXiv:1209.4340*, 2012.